Multimedia Appendix 13. Unified Medical Language System publications for purposes other than the themes noted above.

Author	Publication year	Title	What was UMLS used for?
Auditing			
Goldberg, et al[1]	1998	Validation of clinical problems using a UMLS-based semantic parser	Validation of clinical problems, enrichment of UMLS lexicon, UMLS-based semantic parser
Nawab, et al[2]	2008	Comparing Medline citations using modified N-grams	Identification of duplicate MEDLINE citations
Mahato, et al[3]	2019	A Method to Detect Inconsistent Annotations in a Medical Document using UMLS	Information retrieval, annotated corpus, inconsistency, identification of UMLS concepts
Consumer health			
Leroy, et al[4]	2010	Perils of providing visual health information overviews for consumers with low health literacy or high stress	Visualization, low health literacy or high stress, consumer health
Leroy, et al[5]	2012	Improving perceived and actual text difficulty for health information consumers using semi-automated methods	Simplification of medical text, term familiarity, Easier alternatives from UMLS
Leroy, et al[6]	2013	User evaluation of the effects of a text simplification algorithm using term familiarity on perception, understanding, learning, and information retention	Text simplification, term familiarity, text analysis
Qenam, et al[7]	2017	Text Simplification Using Consumer Health Vocabulary to Generate Patient-Centered Radiology Reporting: Translation and Evaluation	Consumer Health Vocabulary, Radiology Reporting, lexical simplification
Integrated system/data			
Sperzel, et al[8]	1991	Biomedical database inter-connectivity: an experiment linking MIM, GENBANK, and META-1 via MEDLINE	Linking MIM, GENBANK, META-1 via MEDLINE, inter-connectivity
Chute, et al[9]	1994	Medical data and knowledge management by integrated medical workstations: summary and recommendations	Clinical data management, knowledge management, integrated system

Detmer, et al[10]	1997	MedWeaver: integrating decision support, literature searching, and Web exploration using the UMLS Metathesaurus	MedWeaver, decision support, literature search, UMLS
Volot, et al[11]	1997	A UMLS-based method for integrating information databases into an Intranet	Integration of information databases, UMLS-based methods
Joubert, et al[12]	1998	ARIANE: integration of information databases within a hospital intranet	ARIANE, access and query heterogeneous information databases, UMLS
Joubert, et al[13]	1998	Conceptual integration of information databases into an Intranet	ARIANE, access and query information databases, UMLS, conceptual integration
Aymard, et al[14]	2000	Mediation services with health information sources	ARIANE, mediation services, UMLS knowledge sources
Joubert, et al[15]	2001	ARIANE: a mediation framework with health information sources	ARIANE, mediation framework, UMLS, access heterogeneous information sources
Chun, et al[16]	2002	Social health data integration using semantic Web	Link scattered health-related data, RDF
Cantor, et al[17]	2003	Putting data integration into practice: using biomedical terminologies to add structure to existing data sources	Data integration, UMLS, SNOMED CT, semantic relationship, logical relationship
Nardon, et al[18]	2004	Knowledge sharing and information integration in healthcare using ontologies and deductive databases	Ontology, knowledge sharing, information integration
Wang, et al[19]	2004	Automating terminological networks to link heterogeneous biomedical databases	Automated mapping, linking heterogeneous databases
Berlanga, et al[20]	2008	Medical Data Integration and the Semantic Annotation of Medical Protocols	Integration of heterogeneous biomedical information, decision support system
Yang, et al[21]	2008	An integrated database-pipeline system for studying single nucleotide polymorphisms and diseases	SNP and disease, disease and genetic variations, unite disease names and gene names, UMLS
Yamamoto, et al[22]	2011	Building linked open data using approximate string matching methods and domain specific resources	Linked open data set, abbreviation-long form pairs, UMLS as standard term source
Neveol, et al[23]	2012	Linking multiple disease-related resources through UMLS	Multiple disease-related resources, UMLS

Mazouz, et al[24]	2017	Towards a system for integrating heterogeneous health records	Integration of heterogeneous health records, exchange and share patient information
Other research use			
Berman[25]	2003	Concept-match medical data scrubbing. How pathology text can be used in research	Concept match, scrubbing, pathology text, deidentification, UMLS
Zhang, et al[26]	2007	Building user research interest profiles through a MeSH indexer	Research profiles by MeSH terms, MeSH indexer
Boyd, et al[27]	2018	Physician nurse care: A new use of UMLS to measure professional contribution: Are we talking about the same patient a new graph matching algorithm?	UMLS for contribution measurement, graph algorithm

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